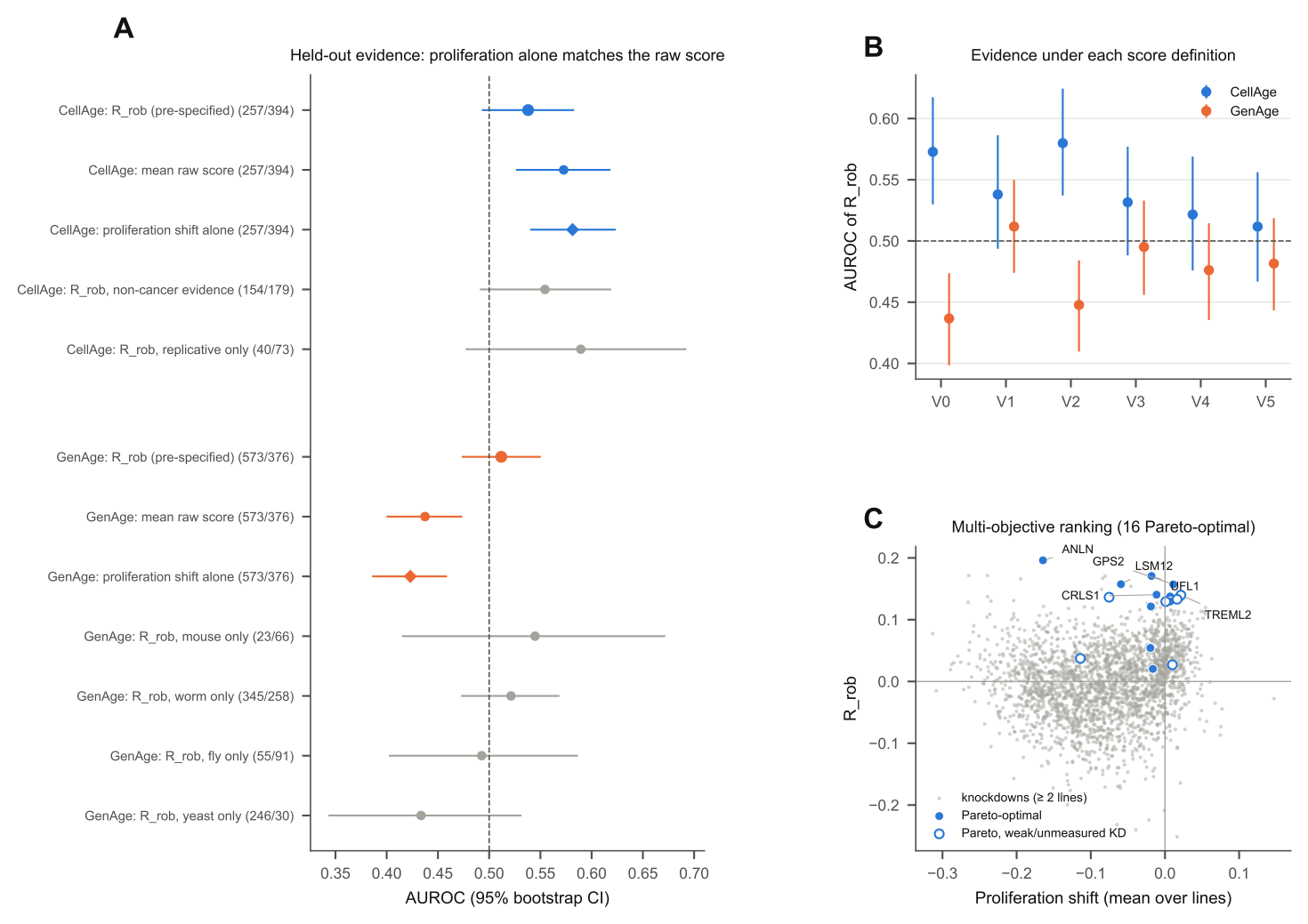

A
Held-out evidence: proliferation alone matches the raw score
CellAge: R_rob (pre-specified) (257/394)
CellAge: mean raw score (257/394)
CellAge: proliferation shift alone (257/394)
CellAge: R_rob, non-cancer evidence (154/179)
CellAge: R_rob, replicative only (40/73)
GenAge: R_rob (pre-specified) (573/376)
GenAge: mean raw score (573/376)
GenAge: proliferation shift alone (573/376)
GenAge: R_rob, mouse only (23/66)
GenAge: R_rob, worm only (345/258)
GenAge: R_rob, fly only (55/91)
GenAge: R_rob, yeast only (246/30)
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
AUROC (95% bootstrap CI)
B
Evidence under each score definition
CellAge
GenAge
AUROC of R_rob
0.60
0.55
0.50
0.45
0.40
V0
V1
V2
V3
V4
V5
C
Multi-objective ranking (16 Pareto-optimal)
ANLN
GPS2
LSM12
UFL1
CRLS1
TREML2
R_rob
0.2
0.1
0.0
−0.1
−0.2
knockdowns (≥ 2 lines)
Pareto-optimal
Pareto, weak/unmeasured KD
−0.3
−0.2
−0.1
0.0
0.1
Proliferation shift (mean over lines)